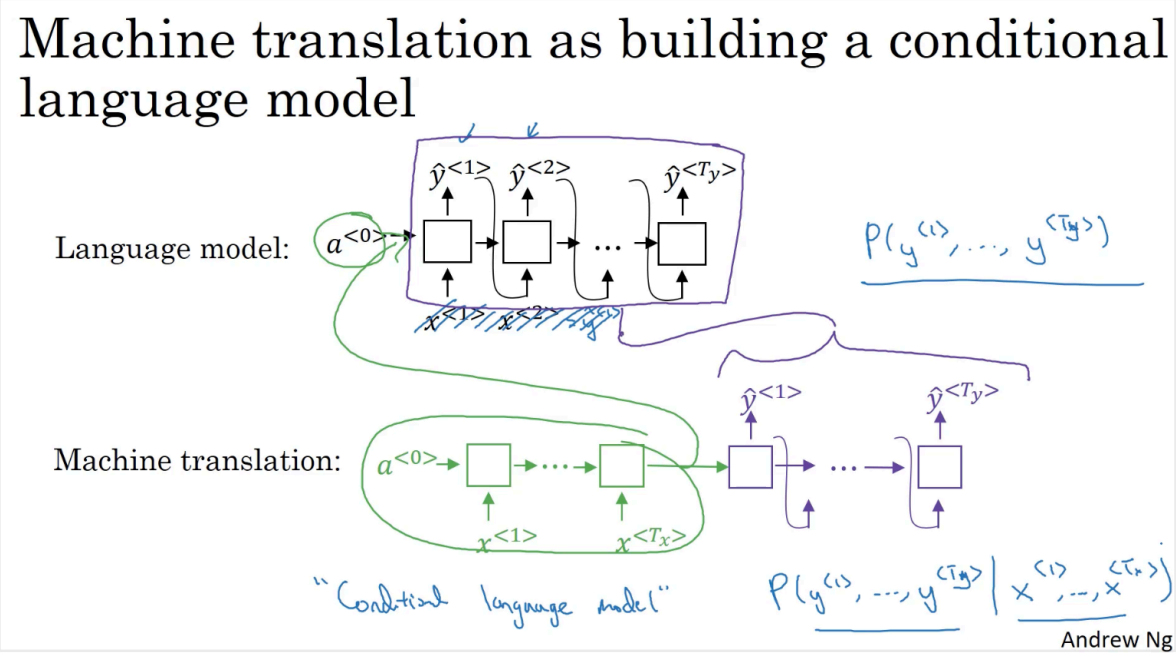
C5W3L02 Picking the most likely sentence

The difference between language model and machine translation is that
1 language model generates a sequence relying on previously generated tokens as well as machine translation model,
2 but what's more important, **language model** takes activations vector *presented* as *zero vector* first, while **machine translation** first tries to figure out some internal representation like encoding and then *uses* this *information* for **generating sequences**.

So that's why machine translation is like building a conditional language model, because we condition on whole sequence



Now we have distribution and we can sample random sequences from it, but we want to find the most suitable one, so we will need algorithm called ***Beam search*** that we will discuss later.

Finding the most likely translation

Jane visite l'Afrique en septembre.

$P(y^{<1>}, \ldots, y^{<T_y>} | x)$

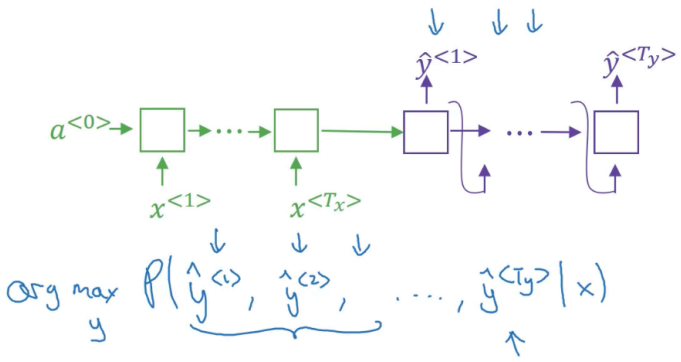
- → Jane is visiting Africa in September.
- → Jane is going to be visiting Africa in September.
- → In September, Jane will visit Africa.
- → Her African friend welcomed Jane in September.

$$\arg\max_{y^{<1>},\ldots,y^{<T_y>}} P(y^{<1>}, \ldots, y^{<T_y>} | x)$$

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Greedy search is not the best option, because it picks the best word just for now and not looks in the future
For example here the probability of taking «going» and not «visiting» may be larger, because «going» is more popular English word, but picking it ends up in more verbose, not that optimal sentence.

Why not a greedy search?



- → Jane is visiting Africa in September.
- → Jane is going to be visiting Africa in September.

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